Use Case Template

CardioAI - Heart Disease Prediction System

1. Use Case Title	Extract Attention Weights
2. Abbreviated Title	Extract Attention
3. Use Case Id	UC-14
4. Actors	ML Engine (Supporting)
5. Description	The ML Engine extracts the learned attention weight distribution from the Attention layer, revealing which clinical features the model focused on most during the heart disease prediction.
5.1. Pre Conditions	1. Model inference (UC-11) has been completed. 2. Attention layer weights are stored during the forward pass.
5.2. Task Sequence	1. System retrieves raw attention scores from the model's attention layer. 2. System applies softmax normalization. 3. System maps each weight to the corresponding clinical feature name. 4. System outputs the attention distribution for visualization.
5.3. Post Conditions	1. A normalized attention weight vector (sums to 1.0) is produced. 2. Weights are mapped to feature names for clinical display.
6. Modification History	Date: September 11, 2026
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